

簡婉軒 Woan-Shiuan Chien (Winnie)

National Yang Ming Chiao Tung University

Department of Electronics and Electrical Engineering

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PROFESSIONAL INTERESTS

Multimodal Learning · Health Analytics · Affective Computing · Trustworthy AI

My research views humans as complex dynamical systems, where internal states such as cognition, perception, emotion, production, and social interaction are expressed through measurable multimodal signals, including speech, language, physiology, gestures, facial expressions, and body movement. By combining signal processing, machine learning, and behavioral science, my work develops computational methods to infer these hidden human states and translate them into human-centered applications in affective computing, healthcare, and interactive AI.

EDUCATION

National Tsing Hua University (NTHU)

PH.D IN DEPARTMENT OF ELECTRICAL ENGINEERING

Dissertation Advisor: Professor [Chi-Chun Lee](#)

From Data Resource Impacts to Fairness Realization in Speech Emotion Recognition

Sep. 2019 -
Mar. 2025

National Chung Cheng University (CCU)

M.S IN DEPARTMENT OF ELECTRICAL ENGINEERING; GPA: 4.02/4.3

Thesis Advisor: Professor [Sung-Nien Yu](#)

Back-propagation and Convolutional Neural Networks for Arrhythmia Electrocardiogram Classification

Sep. 2015 -
Jul. 2017

Chung Yuan Christian University (CYCU)

B.S IN DEPARTMENT OF ELECTRICAL ENGINEERING

Special Research Topic Advisor: Professor [Kang-Ping Lin](#)

Sep. 2011 -
Jun. 2015

PROFESSIONAL EXPERIENCES

Feb. 2026 – Present: Assistant Professor

Department of Electronics and Electrical Engineering, National Yang Ming Chiao Tung University, Hsinchu, Taiwan

May 2025 – Dec. 2025: Postdoctoral Research Fellow

Department of Electrical Engineering, National Tsing Hua University, Hsinchu, Taiwan

Aug. 2019 – Mar. 2025: Graduate Research Assistant

Department of Electrical Engineering, National Tsing Hua University, Hsinchu, Taiwan

Jan. 2024 – Mar. 2024: Visiting Scholar

Intelligent Media Processing Research Team, Artificial Intelligence Research Center (AIRC), National Institute of Advanced Industrial Science and Technology (AIST), Tokyo, Japan

Jul. 2021 – Sep. 2021: Intern

Industrial Technology Research Institute (ITRI), Hsinchu, Taiwan

Oct. 2017 – Sep. 2019: Engineer

Nuvoton Technology Corporation (Nuvoton Corp.), Hsinchu, Taiwan

Aug. 2015 – Jul. 2017: Graduate Research Assistant

Department of Electrical Engineering, National Chung Cheng University, Chiayi, Taiwan

HONORS AND AWARDS

AWARD

IEEE Best Doctoral Dissertation Award, IEEE Taipei Section (2025)

IICM Ph.D. Dissertation Merit Award (2025)

NTHU Outstanding Postdoctoral Research Fellow, National Tsing Hua University (2025)

The Rising Stars Women in Engineering Workshop – Shortlisted Participants (2025) – Shortlisted Candidate (2024)
The 18th TSC Thesis Award: AI Application Competition Finalist, Taiwan Management Institute, Taiwan (2025)
Merry Electronics Co., Ltd.: Electroacoustics Thesis Award Finalist, Taiwan (2024)
Industrial Patent Analysis and Strategy Competition – Second Place, Taiwan Intellectual Property Office, Taiwan (2024)

SCHOLARSHIP

Elite-Well Doctoral Scholarship, Elite-Well Education Foundation, Taiwan (2025 Spring/Fall)
NTHU International Visiting Scholarship, National Tsing Hua University, Taiwan (2024, 2023)
Google Conference Scholarships (APAC), Google (2024, 2023)
NSTC Outstanding Doctoral Students Fellowship, National Science and Technology Council (NSTC), Taiwan (2022-2023)
Rotary Education Scholarship, Chung Hwa Rotary Educational Foundation 2021-2022 (2021-2022)
Academic Excellence Scholarship (Taiwan Life, Cathay Life, Shin Kong Life), Taiwan (2022, 2021, 2016)
Nuvoton Crop. Scholarship, Taiwan (2016-2017)

TRAVEL GRANT

NSTC Outstanding Students Conference Travel Grant, National Science and Technology Council, Taiwan (2025, 2024, 2023, 2022, 2017)
IEEE BSN Travel Awards, IEEE Engineering in Medicine and Biology Society (EMBS) (2024)
ACLCLP Outstanding Students Conference Travel Grant, The Association for Computational Linguistics and Chinese Language Processing (ACLCLP) (2024, 2023)
ACII 2023 Travel Bursary, The Association for the Advancement of Affective Computing (AAAC) (2023)
ICASSP 2023 Conference Travel Grant, IEEE Signal Processing Society (SPS) (2023)
PROGRESS Student Travel Awards, IEEE PROMotInG DiveRsity in Signal ProcESSing (2023)
NTHU Outstanding Students Conference Travel Grant, National Tsing Hua University, Taiwan (2023, 2022)

PUBLICATIONS ([GOOGLE SCHOLAR PROFILE](#), [SCOPUS](#))

Total citations: 322 (H index: 9, i10 index: 9, FWCI: 1.94, Top-1 FWCI: 7.64)

Patent

- [1] Pin-Jhao Chen, **Woan-Shiuan Chien**, Chi-Chun Lee “Contrastive-based Identity Bias Disentanglement Using ADF Detrending”, (*Submit applications for TW, CN and US, April, 2025*)
- [2] Po-Chen Lin, Jeng-Lin Li, **Woan-Shiuan Chien**, Chi-Chun Lee “Method and System for Physiological-based Stress Detection”, (*Submit applications for TW and US, July, 2024*)
- [3] **Woan-Shiuan Chien**, Chi-Chun Lee “Method, model training method and computer program product for speech emotion recognition”, *US Patent App. US18380847 (issued Dec 5, 2022), TW Patent App. TWI859906 (issued Oct 21, 2024) (Submission of applications for CN, May, 2023)*
- [4] **Woan-Shiuan Chien**, “Microcontroller updating system and method”, *CN Patent App. CN112925533A (issued Jun 8, 2021), TW Patent App. TW202123091A (issued Jun 16, 2021)*
- [5] **Woan-Shiuan Chien** and Tzu-Lan Shen, “Recognition system and recognition method”, *US Patent App. US11216729B2 (issued Jan 4, 2022), CN Patent App. CN111292764A (issued Jun 16, 2020), TW Patent App. TWI682325B (issued Jan 11, 2020)*

International Journal

- [1] **Woan-Shiuan Chien** and Chi-Chun Lee, “Only Subsets Matters: The Effect of Dual Fairness Constraints in Speech Emotion Recognition.” in *IEEE Transactions on Affective Computing*, 2026. (Accepted) ***Impact Factor: 9.8 (2024)**
- [2] Luz Martinez-Lucas, Ali Salman, Seong-Gyun Leem, **Woan-Shiuan Chien**, Shreya G. Upadhyay, Chi-Chun Lee and Carlos Busso, “Affective Priming in Emotional Annotations and its Effect on Speech Emotion Recognition” in *IEEE Transactions on Affective Computing*, 2025. (Early Access) ***Impact Factor: 9.8 (2024)**
- [3] **Woan-Shiuan Chien**, Shreya G. Upadhyay, Wei-Cheng Lin, Carlos Busso and Chi-Chun Lee, “Differential Impacts of Monologue and Conversation on Speech Emotion Recognition.” in *IEEE Transactions on Affective Computing*, vol. 16, no. 2, pp. 485–498, 2024. ***Impact Factor: 9.8 (2024)**
- [4] Shreya G. Upadhyay, **Woan-Shiuan Chien**, Bo-Hao Su and Chi-Chun Lee, “Learning with Rater-Expanded Label Space to Improve Speech Emotion Recognition.” in *IEEE Transactions on Affective Computing*, vol. 15, no. 3, pp. 1539–1552, 2024. ***Impact Factor: 9.8 (2024)**
- [5] **Woan-Shiuan Chien** and Chi-Chun Lee, “A Two-Stage Learning Strategy for Fair Speech Emotion Recognition.” in *International Journal of Computational Linguistics and Chinese Language Processing*, vol. 29, no. 1, pp. 1–25, 2024.

International Conference

- [1] **Woan-Shiuan Chien**, Tomohiko Nakamura, Huan-Yu Chen, Satoru Fukayama, Hitoshi Suda, Jun Ogata and Chi-Chun Lee, “Two-Sided Fairness Transfer for Gender-Neutral Speech Emotion Recognition with Partially Observed Attributes.” in *Proceeding of Conference of the International Speech Communication Association (Interspeech ‘26)*, 2026.
- [2] Wei-Heng Huang, **Woan-Shiuan Chien** and Chi-Chun Lee, “Stress Detection Across Daily Activities: A Context-Aware Multimodal Framework with Trajectory and Ambient Speech.” in *Proceeding of Conference of the International Speech Communication Association (Interspeech ‘26)*, 2026.
- [3] Ying-Hsuan Huang, **Woan-Shiuan Chien**, Huan-Yu Chen and Chi-Chun Lee, “Before the Turn: Investigating Motion Cues Preceding Speech in Dyadic Interaction.” in *Proceeding of Conference of the International Speech Communication Association (Interspeech ‘26)*, 2026.

- [4] Huan-Yu Chen, **Woan-Shiuan Chien**, Ching-Heng Lin and Chi-Chun Lee, "Reasoning-driven Tumor Image Synthesis with Anatomical Context." in *Proceeding of 48th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC '26)*, 2026.
- [5] **Woan-Shiuan Chien**, Tomohiko Nakamura, Huan-Yu Chen, Satoru Fukayama, Hitoshi Suda, Jun Ogata and Chi-Chun Lee, "Cross-Attribute Fairness Inference in Speech Emotion Recognition via Task Arithmetic." in *Proceedings of the 159th Spoken Language Information Processing Research Meeting (SIG-SLP '25)*, 2026.
- [6] Huan-Yu Chen, **Woan-Shiuan Chien** and Chi-Chun Lee, "Audience-Aware Co-Speech Gesture Generation in Public Speaking via Anticipation Tokens." in *Proceeding of the 51st IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP '26)*, 2026.
- [7] **Woan-Shiuan Chien**, Huan-Yu Chen, Shu-Ying Liu, Yuan-Liang An, Ching-Shin Hsiao, Yangyang Cai, Victor Pierre Schneider, Miao Cheng, Chia-Huei Tseng and Chi-Chun Lee, "Rethinking Multilingual and Multimodal Interactive Datasets: Challenges, Limitations and Potentials." in *2025 13th IEEE International Conference on Affective Computing and Intelligent Interaction Workshops and Demos (ACIIW '25)*, 2025.
- [8] **Woan-Shiuan Chien**, Mei-Yen Tsai and Chi-Chun Lee, "Graph Structure Learning with Local Connectivity Refinement for Improved Physiological Emotion Recognition." in *Proceeding of 2025 IEEE International Workshop on Machine Learning for Signal Processing (MLSP '25)*, 2025.
- [9] Huan-Yu Chen, **Woan-Shiuan Chien**, Yu-Fen Wang, Bor-Sheng Ko and Chi-Chun Lee, "Instrument-Aware Representation Learning for Robust Cross-Site Flow Cytometry Integration." in *Proceeding of 47th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC '25)*, 2025.
- [10] **Woan-Shiuan Chien**, Huan-Yu Chen and Chi-Chun Lee, "Gradient-Aware Data Augmentation for Federated Stress Detection under Data Incompleteness." in *Proceeding of 47th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC '25)*, 2025.
- [11] Huan-Yu Chen, **Woan-Shiuan Chien**, Ching-Heng Lin and Chi-Chun Lee, "Leveraging Foundation Models for Clinically Instructed Tumor Image Synthesis in Renal Cell Carcinoma." in *Proceeding of the 22nd IEEE International Symposium on Biomedical Imaging (ISBI '25)*, 2025.
- [12] Pin-Jhao Chen, **Woan-Shiuan Chien** and Chi-Chun Lee, "Disentangle Heart Rate Signals for Improved Stress Detection." in *Proceeding of the 50th IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP '25)*, 2025.
- [13] Jing-Chun Wang, **Woan-Shiuan Chien** and Chi-Chun Lee, "A Dynamic Edge-Selection Mechanism in HRV Hypergraph Learning for Improved Stress Detection." in *Proceeding of the 50th IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP '25)*, 2025.
- [14] Shreya G. Upadhyay, **Woan-Shiuan Chien** and Chi-Chun Lee, "Is It Still Fair? Investigating Gender Fairness in Cross-Corpus Speech Emotion Recognition." in *Proceeding of the 50th IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP '25)*, 2025.
- [15] **Woan-Shiuan Chien** and Chi-Chun Lee, "Understanding Missing Data Bias in Longitudinal Mental Stress Detection." in *Proceedings of IEEE 20th International Conference on Body Sensor Networks (BSN '24)*, 2024.
- [16] Pin-Jhao Chen, **Woan-Shiuan Chien**, Huan-Yu Chen and Chi-Chun Lee, "Stress Detection Using HRV Features Augmentation Based on Heart Rate Signal Transformation." in *Proceedings of IEEE 20th International Conference on Body Sensor Networks (BSN '24)*, 2024.
- [17] Jing-Chun Wang, **Woan-Shiuan Chien**, Huan-Yu Chen and Chi-Chun Lee, "In-The-Wild HRV-Based Stress Detection Using Individual-Aware Metric Learning." in *Proceedings of IEEE 20th International Conference on Body Sensor Networks (BSN '24)*, 2024.
- [18] **Woan-Shiuan Chien** and Chi-Chun Lee, "An Investigation of Group versus Individual Fairness in Perceptually Fair Speech Emotion Recognition." in *Proceeding of Conference of the International Speech Communication Association (Interspeech '24)*, 2024, pp. 3205-3209.
- [19] Hsing-Hang Chou, **Woan-Shiuan Chien**, Ya-Tse Wu and Chi-Chun Lee, "An Inter-Speaker Fairness-Aware Speech Emotion Regression Framework." in *Proceeding of Conference of the International Speech Communication Association (Interspeech '24)*, 2024, pp. 3190-3194.
- [20] **Woan-Shiuan Chien**, Shreya G. Upadhyay and Chi-Chun Lee, "Balancing Speaker-Rater Fairness for Gender-Neutral Speech Emotion Recognition." in *Proceeding of the 49th IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP '24)*, 2024, pp. 11861-11865.
- [21] Po-Chen Lin, Jeng-Lin Li, **Woan-Shiuan Chien** and Chi-Chun Lee, "In-the-wild Physiological-based Stress Detection Using Federated Strategy." in *Proceeding of the 49th IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP '24)*, 2024, pp. 1681-1685.
- [22] Shreya G. Upadhyay*, **Woan-Shiuan Chien***, Bo-Hao Su, Lucas Goncalves, Ya-Tse Wu, Ali N. Salman, Carlos Busso and Chi-Chun Lee, "An Intelligent Infrastructure Toward Large Scale Naturalistic Affective Speech Corpora Collection." in *Proceeding of the 11th International Conference on Affective Computing & Intelligent Interaction (ACII '23)*, 2023, pp. 1-8.
- [23] **Woan-Shiuan Chien** and Chi-Chun Lee, "Achieving Fair Speech Emotion Recognition via Perceptual Fairness." in *Proceeding of the 48th IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP '23)*, 2023, pp. 1-5.
- [24] Shreya G. Upadhyay, Luz Martinez-Lucas, Bo-Hao Su, Wei-Cheng Lin, **Woan-Shiuan Chien**, Ya-Tse Wu, William Katz, Carlos Busso and Chi-Chun Lee, "Phonetic Anchor-Based Transfer Learning To Facilitate Unsupervised Cross-Lingual Speech Emotion Recognition." in *Proceeding of the 48th IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP '23)*, 2023, pp. 1-5.
- [25] **Woan-Shiuan Chien**, Shreya G. Upadhyay, Wei-Cheng Lin, Ya-Tse Wu, Bo-Hao Su, Carlos Busso and Chi-Chun Lee, "Monologue versus Conversation: Differences in Emotion Perception and Acoustic Expressivity." in *Proceeding of the 10th IEEE International Conference on Affective Computing & Intelligent Interaction (ACII '22)*, 2022, pp. 1-7.
Invited for Extension in "Best of ACII" of IEEE Transactions on Affective Computing
- [26] **Woan-Shiuan Chien**, Huang-Cheng Chou and Chi-Chun Lee, "Self-assessed Emotion Classification from Acoustic and Physiological Features within Small-group Conversation." in *Companion Publication of the 23rd ACM International Conference on Multimodal Interaction (ICMI '21)*, 2021, pp. 230-239.
- [27] **Woan-Shiuan Chien**, Huang-Cheng Chou and Chi-Chun Lee, "Belongingness and Satisfaction Recognition from Physiological Synchrony with A Group-Modulated Attentive BLSTM under Small-group Conversation." in *Companion Publication of the 23rd ACM International Conference on Multimodal Interaction (ICMI '21)*, 2021, pp. 220-229.

- [28] Huang-Cheng Chou, **Woan-Shiuan Chien**, Da-Chen Juan and Chi-Chun Lee, ““Does it Matter When I Think You Are Lying?” Improving Deception Detection by Integrating Interlocutor’s Judgements in Conversations.” in *Findings of the Joint Conference of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (ACL-IJCNLP ’21)*, 2021, pp. 1846-1860. **Merry Electro-Acoustic Thesis Bronze Award**
- [29] **Woan-Shiuan Chien**, Hao-Chun Yang and Chi-Chun Lee, “Cross Corpus Physiological-based Emotion Recognition Using a Learnable Visual Semantic Graph Convolutional Network.” in *Proceedings of the 25th ACM International Conference on Multimedia (MM ’20)*, 2020, pp. 2999-3006. **Acceptance Rate: 27.8 %**
- [30] **Woan-Shiuan Chien** and Sung-Nien Yu, “Identification of Myocardial Ischemic and Infarction Episodes Based on ST Level and Beat Type Re-attribution Method.” in *Proceedings of the 2nd International Conference on Biomedical Signal and Image Processing (ICBIP ’17)*, 2017, pp. 81-84.

Abstract Paper

- [1] Huan-Yu Chen, **Woan-Shiuan Chien**, Lili Wang, Chi-Chun Lee, Bor-Sheng Ko, Yu-Fen Wang, “Mitigating Inter-Instrument Variability in Flow Cytometry with Sample-Specific Statistical Representations” in *39th annual Congress of the International Society for the Advancement of Cytometry (CYTO ’26)*, Florida, USA, 2026.
- [2] Tzu-Jai Liu, Chao-Jhun Yang, Yi-Chen Lee, **Woan-Shiuan Chien**, Chi-Chun Lee and Yen-Ping Chang, “Clicked! Advanced Quantifications of Couple Emotional Synchrony Using Electrodermal Data and Their Effects on Relationship Quality” in *10th European Conference on Positive Psychology (ECP2 ’22)*, Reykjavik, Iceland, 2022.

SELECTED FUNDED PROJECTS

國科會 National Science and Technology Council (NSTC)

Project Title	Role	Duration	Agency
具包容性之多模態情感互動系統建構：以語音情緒理解為核心的人本訊號處理方法 Human-Centered Signal Processing for Inclusive Multimodal Affective Interaction: Speech Emotion Understanding as the Core	PI	2026/05/01-2028/04/30	NSTC

ACADEMIC SERVICES

COMMITTEE

Organizer of a tutorial on “Understanding and Mitigating Bias in Emotion Recognition Systems” at ACII 2025

INVITED REVIEWER

JOURNAL, IEEE Transactions on Affective Computing (2022-2025), IEEE Transactions on Audio, Speech and Language Processing (2025), Computers in Human Behavior Reports (2025), International Journal of Epidemiology (2024)

CONFERENCE, IEEE ICASSP (2023-2026), IEEE ICME (2025), IEEE ACII (2023-2026), Interspeech (2024, 2023), IEEE BSN (2025), IEEE ISBI (2025)

TEACHING EXPERIENCES

National Yang Ming Chiao Tung University, Taiwan

- EEEC10003 Differential Equations (2026)
- EEEE30049 Governance and Trustworthy Artificial Intelligence (2026)

Future Learn, Massive Open Online Course (MOCC)

TEACHING MATERIAL DESIGNING ASSISTANT

- Deconstructing Research Articles: How to Read and Write a Research Paper (2023, 2024)

National Tsing Hua University, Taiwan

TEACHING ASSISTANT

- THC1024 Reliable Industrial Wireless Network Technology and Application (2023)
- EE 3700 Introduction to Machine Learning (2022)
- EE 3660 Introduction to Digital Signal Processing (2022)
- EE 3900 Special Topic on Implementation (2021, 2022, 2023, 2024)
- EECS 3010 Industry Internship (2021, 2022, 2023, 2024)
- EE 3662 Digital Signal Processing Laboratory (2020)

National Chung Cheng University, Taiwan

TEACHING ASSISTANT

- EE 4156114 Biomedical Signal Processing (2017)
- EE 4151004 Introduction to Computers (2016)
- EE 4153013 Digital Signal Processing Laboratory (2016)
- EE 4153213 Introduction to Digital Signal Processing (2016)

INVITED TALKS

2026/07/08, NTHU Research Talent Resource Center, Hsinchu, Taiwan

Preparing for the Faculty Market: Positioning, Storytelling, and Timing

2026/06/30, Taiwan-France Workshop (IRP e-Light)

AI as the Translation Layer for Heterogeneous Biomedical Sensing

2026/04/17, NTHU EE Graduate Seminar, Hsinchu, Taiwan

AI 聽懂情緒了嗎?從準確率到公平性的情緒語音技術

2026/04/08, NYCU EE Graduate Seminar, Hsinchu, Taiwan

AI 時代下情緒語音技術的發展與公平性挑戰

2026/01/14, Tohoku University, Sendai, Japan

Multimodal Interaction Modeling: Bias-Aware Emotion Understanding in Interactive Contexts

2025/11/17, Rising Star Women in Engineering Workshop 2025, Taipei, Taiwan

Multimodal Signal Processing for Human-Centered AI

2019/08/22, Industrial Technology Research Institute (ITRI), Hsinchu, Taiwan

Nuvoton Microcontroller Development, Implementation and Design for Smart Image Recognition